

# Lab 8 - dataframes con spark

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## Avances

### Toma de datos y fusión

```
In [1]: import pyreadstat
```

```
In [2]: # Spark y utilidades
from pyspark.sql import SparkSession
from pyspark.sql import functions as F
from pyspark.sql import types as T

import pandas as pd
import zipfile, io, re, os, unicodedata
from pathlib import Path

try:
    spark
except NameError:
    spark = SparkSession.builder.appName("AccidentesGT-EDA").getOrCreate()

ZIP_PATH = "./Accidentes.zip"
STAGING_DIR = "./accidentes_staging"
Path(STAGING_DIR).mkdir(parents=True, exist_ok=True)

def normalize_col(col: str) -> str:
    s = unicodedata.normalize('NFKD', str(col))
    s = "".join([c for c in s if not unicodedata.combining(c)])
    s = re.sub(r"^[A-Za-z0-9_]+", "_", s.strip().lower())
    s = re.sub(r"_+", "_", s).strip("_")
    return s

def normalize_df_cols(df: pd.DataFrame) -> pd.DataFrame:
    df = df.copy()
    df.columns = [normalize_col(c) for c in df.columns]
    return df

def read_sav_bytes(b: bytes) -> pd.DataFrame:
    bio = io.BytesIO(b)
    try:
        pdf = pd.read_spss(bio, convert_categoricals=False)
        return pdf
    except Exception:
```

```

        pass
    try:
        pdf, meta = pyreadstat.read_sav(io.BytesIO(b))
        return pdf
    except Exception as e:
        raise RuntimeError("No se pudo leer el .sav; instala 'pyreadstat'") from e

```

In [ ]: *# Ruta temporal para Los sav (error con windows)*

```

import os, io, re, unicodedata, zipfile, tempfile, pathlib
import pandas as pd

try:
    import pyreadstat
    print("pyreadstat OK - versión:", pyreadstat.__version__)
except Exception as e:
    print("pyreadstat no se pudo importar en este kernel:", repr(e))

def read_sav_bytes(b: bytes) -> pd.DataFrame:
    with tempfile.NamedTemporaryFile(delete=False, suffix=".sav") as tmp:
        tmp.write(b)
        tmp_path = tmp.name

    try:
        try:
            pdf = pd.read_spss(tmp_path, convert_categoricals=False)
            return pdf
        except Exception as e1:
            try:
                import pyreadstat
                pdf, meta = pyreadstat.read_sav(tmp_path, apply_value_formats=False)
                return pdf
            except Exception as e2:
                raise RuntimeError(f"Fallo leyendo .sav; pandas_error={repr(e1)} | {repr(e2)}")
    finally:
        try:
            os.unlink(tmp_path)
        except Exception:
            pass

print("Hotfix read_sav_bytes cargado.")

```

pyreadstat OK - versión: 1.2.7

Hotfix read\_sav\_bytes cargado.

```

In [ ]: import numpy as np
import pandas as pd

NUM_REGEX = r"^[+-]?(\d+([.]\d+)?)|\. \d+)$"

def smart_cast_dataframe(df: pd.DataFrame, numeric_threshold: float = 0.95) -> pd.DataFrame:
    df = df.copy()

    NULL_LIKE = {"", "nan", "none", "sin dato", "s/d", "na", "n/a", "null"}
    for c in df.columns:
        if df[c].dtype == object:

```

```

s = df[c].astype(str).str.strip()
s = s.where(~s.str.lower().isin(NULL_LIKE), other=pd.NA)
df[c] = s

for c in df.columns:
    if pd.api.types.is_string_dtype(df[c]) or df[c].dtype == object:
        s = df[c]
        s_dot = s.str.replace(".", ".", regex=False)
        mask_num = s_dot.fillna("").str.match(NUM_REGEX)
        denom = (s_dot.notna()).sum()
        ratio = (mask_num.sum() / denom) if denom else 0.0

        if ratio >= numeric_threshold and denom > 0:
            df[c] = pd.to_numeric(s_dot, errors="coerce")
        else:
            try:
                df[c] = s.astype("string[pyarrow]")
            except Exception:
                df[c] = s.astype("string")

for c in df.columns:
    if pd.api.types.is_float_dtype(df[c]):
        s = df[c].dropna()
        if len(s) and np.all(np.isclose(s, np.round(s))):
            df[c] = df[c].round().astype("Int64")

try:
    df = df.convert_dtypes(dtype_backend="pyarrow")
except Exception:
    pass

return df

```

```

In [ ]: BUCKETS = {
    "hechos": re.compile(r"hecho|incidente|accidente|hechos", re.I),
    "vehiculos": re.compile(r"vehicul", re.I),
    "fallecidos": re.compile(r"fallecid|muert", re.I),
}

dfs_acc = {k: [] for k in BUCKETS.keys()}
bucket_files = {k: [] for k in BUCKETS.keys()}

# Extrae el zip

with zipfile.ZipFile(ZIP_PATH, "r") as z:
    members = z.namelist()
    data_files = [n for n in members if n.lower().endswith((".sav", ".sas7bdat", ".
    for fname in data_files:
        bkt = None
        base = Path(fname).name
        for k, rx in BUCKETS.items():
            if rx.search(base):
                bkt = k
                break
        if bkt is None:
            continue

```

```

with z.open(fname) as f:
    raw = f.read()

ext = base.lower().split(".")[-1]
try:
    if ext == "sav":
        pdf = read_sav_bytes(raw)
    elif ext in ("xls", "xlsx"):
        xls = pd.ExcelFile(io.BytesIO(raw))
        for sh in xls.sheet_names:
            try:
                tmp = pd.read_excel(io.BytesIO(raw), sheet_name=sh, dtype=s
                if tmp.shape[0] == 0:
                    continue
                tmp = normalize_df_cols(tmp)
                dfs_acc[bkt].append(tmp)
                bucket_files[bkt].append(f"{base}::{sh}")
            except Exception:
                pass
            continue
        else:
            continue

    if isinstance(pdf, pd.DataFrame) and pdf.shape[0] > 0:
        pdf = normalize_df_cols(pdf)
        dfs_acc[bkt].append(pdf)
        bucket_files[bkt].append(base)
except Exception as e:
    print(f"No se pudo leer {base}: {e}")

for k in dfs_acc.keys():
    print(f"\n=== {k.upper()} ===")
    for it in bucket_files[k]:
        print(" -", it)

# Combina por bucket y guarda como parquet
combined_paths = {}
for k, lst in dfs_acc.items():
    if not lst:
        continue
    combo = pd.concat(lst, ignore_index=True)
    combo = combo.dropna(axis=1, how="all")

    combo = smart_cast_dataframe(combo, numeric_threshold=0.95)

    out_path = f"{STAGING_DIR}/{k}_pandas.parquet"
    combo.to_parquet(out_path, index=False)
    combined_paths[k] = out_path

combined_paths

```

```

=== HECHOS ===
- 2013 hechos.sav
- 2014 hechos.sav
- 2015 hechos.xlsx::Eventos de tránsito
- 2016 hechos.sav
- 2017 hechos.sav
- 2018 hechos.sav
- 2019 hechos.sav
- 2020 hechos.sav
- 2021 hechos.sav
- 2022 hechos.sav
- 2023 hechos.sav

=== VEHICULOS ===
- 2013 vehiculos.sav
- 2014 vehiculos.sav
- 2015 vehiculos.sav
- 2015 vehiculos.xlsx::Vehículos involucrados
- 2017 vehiculos.sav
- 2018 vehiculos.sav
- 2019 vehiculos.sav
- 2020 vehiculos.sav
- 2021 vehiculos.sav
- 2022 vehiculos.sav
- 2023 vehiculos.sav

=== FALLECIDOS ===
- 2013 fallecidos.sav
- 2014 fallecidos.sav
- 2015 fallecidos.xlsx::Fallecidos y Lesionados
- 2016 fallecidos.sav
- 2017 fallecidos.sav
- 2018 fallecidos.sav
- 2019 fallecidos.sav
- 2020 fallecidos.sav
- 2021 fallecidos.sav
- 2022 fallecidos.sav
- 2023 fallecidos.sav

```

```

Out[ ]: {'hechos': './accidentes_staging/hechos_pandas.parquet',
        'vehiculos': './accidentes_staging/vehiculos_pandas.parquet',
        'fallecidos': './accidentes_staging/fallecidos_pandas.parquet'}

```

```

In [ ]: sdf = {}
        for name, path in combined_paths.items():
            df = spark.read.parquet(path)
            for c in df.columns:
                df = df.withColumnRenamed(c, normalize_col(c))
            sdf[name] = df.cache()

        { k: (v.count(), len(v.columns)) for k, v in sdf.items() }

```

```

Out[ ]: {'hechos': (76465, 38), 'vehiculos': (112030, 38), 'fallecidos': (108941, 41)}

```

## Estandarizacion de nombres

```

In [ ]: from pyspark.sql import functions as F

ALIASES = {
    "anio": [
        "anio", "ano", "year", "ano_",
        "anio_ocu", "ano_ocu", "year_ocu"
    ],
    "mes": [
        "mes", "mes_num", "mesnumero", "mes_", "mes_nombre",
        "mes_ocu"
    ],
    "departamento": [
        "departamento", "depto", "depto_", "departament", "departamento_",
        "depto_ocu", "departamento_ocu"
    ],
    "tipo_accidente": [
        "tipo_accidente", "tipo", "clase_accidente",
        "tipo_hecho", "clase_hecho", "evento", "tipo_evento",
        "clase_acc", "causa_acc"
    ],
}

MES_MAP = {
    "ene":1,"enero":1,"jan":1,"january":1,
    "feb":2,"febrero":2,"february":2,
    "mar":3,"marzo":3,"march":3,
    "abr":4,"abril":4,"apr":4,"april":4,
    "may":5,"mayo":5,
    "jun":6,"junio":6,
    "jul":7,"julio":7,
    "ago":8,"agosto":8,"aug":8,"august":8,
    "sep":9,"set":9,"sept":9,"septiembre":9,"september":9,
    "oct":10,"octubre":10,"oct":10,"october":10,
    "nov":11,"noviembre":11,"november":11,
    "dic":12,"diciembre":12,"dec":12,"december":12,
}

def coalesce_col(df, target, candidates):
    cols = [c for c in candidates if c in df.columns]
    if not cols:
        return df, None
    expr = None
    for c in cols:
        expr = F.coalesce(expr, F.col(c)) if expr is not None else F.col(c)
    return df.withColumn(target, expr), target

def ensure_keys(df):
    for target, cands in ALIASES.items():
        df, _ = coalesce_col(df, target, cands)

    if "anio" in df.columns:
        df = df.withColumn("anio", F.col("anio").cast("int"))

    if "mes" in df.columns:
        df = df.withColumn("mes", F.col("mes").cast("int"))

```

```

mes_expr = F.lower(F.regexp_replace(F.coalesce(F.col("mes").cast("string"),
mes_map_lit = F.create_map([F.lit(k) for kv in MES_MAP.items() for k in kv]
df = df.withColumn(
    "mes",
    F.when(F.col("mes").isNull(), mes_map_lit.getItem(mes_expr))
      .otherwise(F.col("mes")).cast("int")
)

if "departamento" in df.columns:
    df = df.withColumn("departamento", F.trim(F.initcap(F.col("departamento"))))

if "tipo_accidente" in df.columns:
    clean = F.regexp_replace(F.lower(F.col("tipo_accidente")), r"\s+", " ")
    df = df.withColumn("tipo_accidente", F.trim(clean))
    df = (df
        .withColumn("tipo_accidente",
            F.when(F.col("tipo_accidente").isin("colision", "colisión"), F.lit("
            .when(F.col("tipo_accidente").isin("atropello", "atropellamiento"),
            .otherwise(F.col("tipo_accidente")))
        )
    )

return df

for k in list(sdf.keys()):
    sdf[k] = ensure_keys(sdf[k]).cache()

for k, df in sdf.items():
    have = {c for c in ["anio", "mes", "departamento", "tipo_accidente"] if c in df.co
    print(f"{k}: claves presentes -> {sorted(have)}")

```

C:\Users\sebas\AppData\Local\Packages\PythonSoftwareFoundation.Python.3.9\_qbz5n2kfra8p0\LocalCache\local-packages\Python39\site-packages\pyspark\sql\classic\column.py:359: FutureWarning: A column as 'key' in getItem is deprecated as of Spark 3.0, and will not be supported in the future release. Use `column[key]` or `column.key` syntax instead.

```
warnings.warn(
```

hechos: claves presentes -> ['anio', 'departamento', 'mes', 'tipo\_accidente']

vehiculos: claves presentes -> ['anio', 'departamento', 'mes', 'tipo\_accidente']

fallecidos: claves presentes -> ['anio', 'departamento', 'mes', 'tipo\_accidente']

## Conteo de datos

```

In [ ]: for k, df in sdf.items():
        print(f"=== {k.upper()} ===")
        print("Filas:", df.count(), " | Columnas:", len(df.columns))
        df.show(10, truncate=False)

DESCRIBE_COLS = {
    "hechos": ["anio", "mes", "departamento", "tipo_accidente"],
    "vehiculos": ["anio", "mes", "departamento", "tipo_accidente"],
    "fallecidos": ["anio", "mes", "departamento", "tipo_accidente"],
    "lesionados": ["anio", "mes", "departamento", "tipo_accidente"],
}

```

```
for k, df in sdf.items():
    print(f"\n--- Describe: {k} ---")
    cols = [c for c in DESCRIBE_COLS.get(k, []) if c in df.columns]
    if cols:
        df.select(*cols).describe().show(truncate=False)
        num_cols = [c for c, t in df.dtypes if t in ("int", "bigint", "double", "fl
        if num_cols:
            df.select(*num_cols).summary().show(truncate=False)
```



=== HECHOS ===

Filas: 76465 | Columnas: 42

```

+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
|num_hecho|dia_ocu|mes_ocu|dia_sem_ocu|hora_ocu|g_hora|depto_ocu|mupio_ocu|areag_ocu
|zona_ocu|sexo_pil|edad_pil|g_edad_2|mayor_menor|tipo_veh|color_veh|modelo_veh|causa
_acc|marca_veh|estado_pil|num_correlativo|corre_base|area_geo_ocu|sexo_con|edad_con|
g_edad|estado_con|tipo_eve|num_corre|ano_ocu|g_hora_5|sexo_per|edad_per|g_edad_80yma
s|g_edad_60ymas|edad_quinquenales|g_modelo_veh|zona_ciudad|anio|mes|departamento|tip
o_accidente|
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
|1        |1        |1        |2        |3        |1        |1        |106       |1
|6        |1        |32       |5        |1        |3        |99       |9999      |2
|99       |2        |NULL     |NULL     |NULL     |NULL     |NULL     |NULL     |NULL
|NULL     |NULL     |NULL     |NULL     |NULL     |NULL     |NULL     |NULL     |NULL
|NULL     |NULL     |NULL     |NULL     |NULL     |1        |1        |2        |
|2        |1        |1        |2        |5        |1        |1        |101       |1
|11       |1        |34       |5        |1        |1        |2        |2010      |2
|5        |2        |NULL     |NULL     |NULL     |NULL     |NULL     |NULL     |NULL
|NULL     |NULL     |NULL     |NULL     |NULL     |NULL     |NULL     |NULL     |NULL
|NULL     |NULL     |NULL     |NULL     |NULL     |1        |1        |2        |
|3        |1        |1        |2        |3        |1        |1        |108       |1
|3        |1        |28       |4        |1        |1        |4        |9999      |1
|20       |2        |NULL     |NULL     |NULL     |NULL     |NULL     |NULL     |NULL
|NULL     |NULL     |NULL     |NULL     |NULL     |NULL     |NULL     |NULL     |NULL
|NULL     |NULL     |NULL     |NULL     |NULL     |1        |1        |1        |
|4        |1        |1        |2        |0        |1        |19       |1901      |2
|99       |1        |21       |3        |1        |4        |1        |2012      |2
|24       |1        |NULL     |NULL     |NULL     |NULL     |NULL     |NULL     |NULL
|NULL     |NULL     |NULL     |NULL     |NULL     |NULL     |NULL     |NULL     |NULL
|NULL     |NULL     |NULL     |NULL     |NULL     |1        |19       |2        |
|5        |1        |1        |2        |18       |4        |19       |1901      |2
|99       |1        |19       |2        |1        |3        |1        |9999      |3
|44       |9        |NULL     |NULL     |NULL     |NULL     |NULL     |NULL     |NULL
|NULL     |NULL     |NULL     |NULL     |NULL     |NULL     |NULL     |NULL     |NULL
|NULL     |NULL     |NULL     |NULL     |NULL     |1        |19       |3        |
|6        |1        |1        |2        |0        |1        |5        |502       |2
|99       |1        |24       |3        |1        |4        |5        |2001      |2
|2        |1        |NULL     |NULL     |NULL     |NULL     |NULL     |NULL     |NULL
|NULL     |NULL     |NULL     |NULL     |NULL     |NULL     |NULL     |NULL     |NULL
|NULL     |NULL     |NULL     |NULL     |NULL     |1        |5        |2        |
|7        |1        |1        |2        |11       |2        |9        |901       |2
|99       |1        |36       |6        |1        |2        |5        |9999      |2
|44       |2        |NULL     |NULL     |NULL     |NULL     |NULL     |NULL     |NULL
|NULL     |NULL     |NULL     |NULL     |NULL     |NULL     |NULL     |NULL     |NULL
|NULL     |NULL     |NULL     |NULL     |NULL     |1        |9        |2        |
|8        |1        |1        |2        |1        |1        |12       |1217      |2

```

|      |      |      |      |      |      |      |      |      |
|------|------|------|------|------|------|------|------|------|
| 99   | 1    | 38   | 6    | 1    | 5    | 6    | 1992 | 1    |
| 34   | 1    | NULL |      | NULL | NULL |      | NULL | NULL |
| NULL | NULL | NULL | NULL | NULL | NULL | NULL | NULL | NULL |
| NULL |      | NULL | NULL | NULL | 1    | 12   | 1    |      |
| 9    | 1    | 1    | 2    | 7    | 2    | 13   | 1322 | 2    |
| 99   | 9    | 999  | 12   | 1    | 3    | 99   | 9999 | 3    |
| 44   | 1    | NULL |      | NULL | NULL |      | NULL | NULL |
| NULL | NULL | NULL | NULL | NULL | NULL | NULL | NULL | NULL |
| NULL |      | NULL | NULL | NULL | 1    | 13   | 3    |      |
| 10   | 1    | 1    | 2    | 8    | 2    | 16   | 1604 | 2    |
| 99   | 1    | 33   | 5    | 1    | 5    | 2    | 1999 | 2    |
| 19   | 2    | NULL |      | NULL | NULL |      | NULL | NULL |
| NULL | NULL | NULL | NULL | NULL | NULL | NULL | NULL | NULL |
| NULL |      | NULL | NULL | NULL | 1    | 16   | 2    |      |

only showing top 10 rows

=== VEHICULOS ===

Filas: 112030 | Columnas: 42

```
|num_hecho|dia_ocu|mes_ocu|dia_sem_ocu|hora_ocu|depto_ocu|mupio_ocu|areag_ocu|zona_o  
cu|sexo_pil|edad_pil|mayor_menor|tipo_veh|color_veh|modelo_veh|causa_acc|marca_veh|g  
_edad|estado_pil|grupo_mode_veh|num_correlativo|corre_base|g_hora|area_geo_ocu|sexo_  
con|edad_con|estado_con|tipo_eve|num_corre|ano_ocu|sexo_per|edad_per|g_edad_80ymas|g  
_edad_60ymas|edad_quinquenales|g_hora_5|g_modelo_veh|zona_ciudad|anio|mes|departamen  
to|tipo_accidente|
```

|      |      |      |      |      |      |      |      |      |
|------|------|------|------|------|------|------|------|------|
| 1    | 1    | 1    | 2    | 3    | 1    | 106  | 1    | 6    |
| 1    | 32   | 1    | 3    | 99   | 6    | 2    | 99   | 5    |
| 2    | 6    |      | NULL |      | NULL | NULL | NULL | N    |
| ULL  | NULL | NULL | NULL | NULL | NULL | NULL | NULL | NULL |
| NULL |      | NULL | NULL | NULL | NULL | 1    | 1    | 2    |
|      |      |      |      |      |      |      |      |      |
| 2    | 1    | 1    | 2    | 5    | 1    | 101  | 1    | 11   |
| 1    | 34   | 1    | 1    | 2    | 5    | 2    | 5    | 5    |
| 2    | 5    |      | NULL |      | NULL | NULL | NULL | N    |
| ULL  | NULL | NULL | NULL | NULL | NULL | NULL | NULL | NULL |
| NULL |      | NULL | NULL | NULL | NULL | 1    | 1    | 2    |
|      |      |      |      |      |      |      |      |      |
| 3    | 1    | 1    | 2    | 3    | 1    | 108  | 1    | 3    |
| 1    | 28   | 1    | 1    | 4    | 6    | 1    | 20   | 4    |
| 2    | 6    |      | NULL |      | NULL | NULL | NULL | N    |

|      |      |      |      |      |      |      |      |      |
|------|------|------|------|------|------|------|------|------|
| ULL  | NULL | NULL | NULL | NULL | NULL | NULL | NULL | NULL |
| NULL |      | NULL | NULL | NULL | NULL | 1    | 1    | 1    |
|      |      |      |      |      |      |      |      |      |
| 4    | 1    | 1    | 2    | 3    | 1    | 108  | 1    | 3    |
| 1    | 43   | 1    | 3    | 6    | 4    | 1    | 44   | 7    |
| 1    | 4    |      | NULL | NULL | NULL | NULL | NULL | N    |
| ULL  | NULL | NULL | NULL | NULL | NULL | NULL | NULL | NULL |
| NULL |      | NULL | NULL | NULL | NULL | 1    | 1    | 1    |
|      |      |      |      |      |      |      |      |      |
| 5    | 1    | 1    | 2    | 0    | 19   | 1901 | 2    | 99   |
| 1    | 21   | 1    | 4    | 1    | 5    | 2    | 24   | 3    |
| 1    | 5    |      | NULL | NULL | NULL | NULL | NULL | N    |
| ULL  | NULL | NULL | NULL | NULL | NULL | NULL | NULL | NULL |
| NULL |      | NULL | NULL | NULL | NULL | 1    | 19   | 2    |
|      |      |      |      |      |      |      |      |      |
| 6    | 1    | 1    | 2    | 18   | 19   | 1901 | 2    | 99   |
| 1    | 19   | 1    | 3    | 1    | 6    | 3    | 44   | 2    |
| 9    | 6    |      | NULL | NULL | NULL | NULL | NULL | N    |
| ULL  | NULL | NULL | NULL | NULL | NULL | NULL | NULL | NULL |
| NULL |      | NULL | NULL | NULL | NULL | 1    | 19   | 3    |
|      |      |      |      |      |      |      |      |      |
| 7    | 1    | 1    | 2    | 0    | 5    | 502  | 2    | 99   |
| 1    | 24   | 1    | 4    | 5    | 4    | 2    | 2    | 3    |
| 1    | 4    |      | NULL | NULL | NULL | NULL | NULL | N    |
| ULL  | NULL | NULL | NULL | NULL | NULL | NULL | NULL | NULL |
| NULL |      | NULL | NULL | NULL | NULL | 1    | 5    | 2    |
|      |      |      |      |      |      |      |      |      |
| 8    | 1    | 1    | 2    | 11   | 9    | 901  | 2    | 99   |
| 1    | 36   | 1    | 2    | 5    | 6    | 2    | 44   | 6    |
| 2    | 6    |      | NULL | NULL | NULL | NULL | NULL | N    |
| ULL  | NULL | NULL | NULL | NULL | NULL | NULL | NULL | NULL |
| NULL |      | NULL | NULL | NULL | NULL | 1    | 9    | 2    |
|      |      |      |      |      |      |      |      |      |
| 9    | 1    | 1    | 2    | 1    | 12   | 1217 | 2    | 99   |
| 1    | 38   | 1    | 5    | 6    | 3    | 1    | 34   | 6    |
| 1    | 3    |      | NULL | NULL | NULL | NULL | NULL | N    |
| ULL  | NULL | NULL | NULL | NULL | NULL | NULL | NULL | NULL |
| NULL |      | NULL | NULL | NULL | NULL | 1    | 12   | 1    |
|      |      |      |      |      |      |      |      |      |
| 10   | 1    | 1    | 2    | 1    | 12   | 1217 | 2    | 99   |
| 9    | 999  | 1    | 4    | 3    | 6    | 1    | 40   | 11   |
| 2    | 6    |      | NULL | NULL | NULL | NULL | NULL | N    |
| ULL  | NULL | NULL | NULL | NULL | NULL | NULL | NULL | NULL |
| NULL |      | NULL | NULL | NULL | NULL | 1    | 12   | 1    |
|      |      |      |      |      |      |      |      |      |

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only showing top 10 rows
=== FALLECIDOS ===
Filas: 108941 | Columnas: 45
+-----+-----+-----+-----+-----+-----+-----+-----+-----+
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+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
|num_hecho|dia_ocu|mes_ocu|dia_sem_ocu|hora_ocu|g_hora|depto_ocu|mupio_ocu|areag_ocu
|zona_ocu|sexo_pil|edad_pil|g_edad_2|mayor_menor|tipo_veh|color_veh|modelo_veh|causa
_acc|marca_veh|g_edad|fallecidos_lesionados|otro_g_edad_fall_les|num_correlativo|num
_corre|corre_base|area_geo_ocu|sexo_vic|edad_vic|fall_les|tipo_eve|edad_quinquenales
|ano_ocu|g_hora_5|sexo_per|edad_per|g_edad_80ymas|g_edad_60ymas|int_o_noint|g_modelo
_veh|filter|zona_ciudad|anio|mes|departamento|tipo_accidente|
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
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+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
|1|1|1|2|5|1|1|101|1
|11|1|34|5|1|1|2|5|2
|5|5|2|4|NULL|NULL
|NULL|NULL|NULL|NULL|NULL|NULL|NULL|NULL|NULL
|NULL|NULL|NULL|NULL|NULL|NULL|NULL|NULL|NULL
L|NULL|NULL|1|1|2|1|101|1
|2|1|1|2|5|1|1|101|1
|11|2|15|4|2|1|2|5|2
|5|2|2|3|NULL|NULL
|NULL|NULL|NULL|NULL|NULL|NULL|NULL|NULL|NULL
|NULL|NULL|NULL|NULL|NULL|NULL|NULL|NULL|NULL
L|NULL|NULL|1|1|2|1|101|1
|3|1|1|2|5|1|1|101|1
|11|2|34|5|1|1|2|5|2
|5|5|2|4|NULL|NULL
|NULL|NULL|NULL|NULL|NULL|NULL|NULL|NULL|NULL
|NULL|NULL|NULL|NULL|NULL|NULL|NULL|NULL|NULL
L|NULL|NULL|1|1|2|1|101|1
|4|1|1|2|0|1|19|1901|2
|99|1|21|4|1|4|1|5|2
|24|3|1|3|NULL|NULL
|NULL|NULL|NULL|NULL|NULL|NULL|NULL|NULL|NULL
|NULL|NULL|NULL|NULL|NULL|NULL|NULL|NULL|NULL
L|NULL|NULL|1|19|2|1|19|1901|2
|5|1|1|2|0|1|19|1901|2
|99|1|22|4|1|4|1|5|2
|24|3|2|3|NULL|NULL
|NULL|NULL|NULL|NULL|NULL|NULL|NULL|NULL|NULL
|NULL|NULL|NULL|NULL|NULL|NULL|NULL|NULL|NULL
L|NULL|NULL|1|19|2|1|19|1901|2
|6|1|1|2|18|4|19|1901|2
|99|1|19|4|1|3|1|6|3
|44|2|2|3|NULL|NULL
|NULL|NULL|NULL|NULL|NULL|NULL|NULL|NULL|NULL
|NULL|NULL|NULL|NULL|NULL|NULL|NULL|NULL|NULL
L|NULL|NULL|1|19|3|1|19|1901|2
|7|1|1|2|18|4|19|1901|2
|99|1|11|4|2|3|1|6|3
|44|1|2|2|NULL|NULL
|NULL|NULL|NULL|NULL|NULL|NULL|NULL|NULL|NULL

```

--- Describe: hechos ---

13/24

14/24

--- Describe: vehiculos ---

15/24

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-----+-----+-----+-----+-----+-----+
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+-----+-----+-----+-----+-----+-----+
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|summary|num_hecho      |dia_ocu      |mes_ocu      |dia_sem_ocu  |h
ora_ocu      |depto_ocu      |mupio_ocu      |areag_ocu      |zona_ocu
|sexo_pil      |edad_pil      |mayor_menor      |tipo_veh      |color_ve
h      |modelo_veh      |causa_acc      |marca_veh      |g_edad
|estado_pil      |grupo_mode_veh |num_correlativo |corre_base      |g_hora
|area_geo_ocu      |sexo_con      |edad_con      |estado_con      |tipo_e
ve      |num_corre      |ano_ocu      |sexo_per      |edad_per
|g_edad_80ymas      |g_edad_60ymas      |edad_quinquenales|g_hora_5      |g_modelo_
veh      |zona_ciudad      |anio      |mes      |
+-----+-----+-----+-----+-----+-----+
-----+-----+-----+-----+-----+-----+
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--+
|count |6323      |112030      |112030      |112030      |1
12030      |112030      |112030      |6323      |112030
|6323      |6323      |112030      |112030      |112030
|112030      |6323      |112030      |14227      |6323
|6323      |7904      |7904      |105707      |29345
|7904      |7904      |105707      |105707      |97803
|87980      |97803      |97803      |97803      |97803
|97803      |97803      |97803      |12796      |87980
|112030      |
|mean |3162.0      |15.65995715433366|6.613175042399357 |4.347478354012318|1
3.705712755511916|7.167169508167455|722.2372757297153|1.6212241024829985 |71.7415602
9634919|1.4206863830460223|176.7659338921398|2.301981612068196 |9.03912344907614 |2
9.399312684102473|6282.1339730429345|1.4445674521587855|258.81995001338925|6.6159415
19645744|5.027202277399969|5.002214138858137 |3952.5      |2331.886639676113
|2.8834230467234905|1.5937979212813085 |2.6052631578947367|298.6318319838057 |6.4568
95002223126 |2.1765256794725043|5518.339846425979|2019.5395885428507|2.0442624459372
41 |258.141457828492|7.701778064067565|6.744138727850884 |9.701655368444731|2.029048
188705868 |47.260523705816794|80.28477649265396 |2019.5395885428507|6.61317504239935
7 |
|stddev |1825.4372079039038|8.79666485589112 |3.5230422763393103|2.041437925562715|
6.978650029801068 |6.898342983301385|689.6747997127318|0.48512053858575366|41.852351
59745763|1.653128437821971 |343.9991515365342|2.9251503646441317|21.841041318964585|
42.20492728156827 |4292.583815291279 |3.9326286373575132|398.361181950327 |3.347168
369428716|3.8902070693114483|1.3269628011193737|2281.8325968396543|1606.250294445282
2|1.08262002971575 |0.49113151982746434|3.1472452409906135|431.27241632289855|3.653
0313927166156|2.2311465179591243|3277.609318707572|2.144553604790348 |2.613747132926
4512|408.693338610941|5.046231897040685|3.5467981009118783|5.046409026673938|0.85381
98547516217|46.822669986382465|36.362129699947594|2.144553604790348 |3.5230422763393
103|
|min |1      |1      |1      |1      |0

```



|         |         |         |         |         |
|---------|---------|---------|---------|---------|
| 1       | 101     | 1       | 1       | 1       |
| 7       | 1       | 1       | 1       | 1       |
| 1       | 1       | 1       | 1       | 1       |
| 1       | 1       | 1       | 1       | 1       |
| 12      | 1       | 1       | 1       | 2016    |
| 1       | 3       | 1       | 1       | 1       |
| 1       | 1       | 1       | 2016    | 1       |
|         |         |         |         |         |
| 25%     | 1580    | 8       | 3       | 3       |
| 1       | 108     | 1       | 13      | 1       |
| 25      | 1       | 2       | 2       | 2006    |
| 1       | 27      | 4       | 1       | 4       |
| 1975    | 988     | 2       | 1       | 1       |
| 26      | 1       | 1       | 2718    | 2018    |
| 1       | 25      | 4       | 4       | 6       |
| 1       | 5       | 99      | 2018    | 3       |
|         |         |         |         |         |
| 50%     | 3161    | 16      | 7       | 5       |
| 5       | 5       | 501     | 2       | 99      |
| 1       | 33      | 1       | 4       | 5       |
| 9999    | 1       | 56      | 6       | 2       |
| 6       | 3951    | 1976    | 3       | 2       |
| 1       | 37      | 9       | 1       | 5434    |
| 2020    | 1       | 35      | 6       | 6       |
| 8       | 2       | 6       | 99      | 2020    |
| 7       |         |         |         |         |
| 75%     | 4742    | 23      | 10      | 6       |
| 9       | 13      | 1301    | 2       | 99      |
| 1       | 47      | 1       | 4       | 99      |
| 9999    | 1       | 99      | 10      | 9       |
| 6       | 5927    | 3651    | 4       | 2       |
| 1       | 999     | 9       | 4       | 8150    |
| 2022    | 1       | 64      | 11      | 11      |
| 13      | 3       | 99      | 99      | 2022    |
| 10      |         |         |         |         |
| max     | 6323    | 31      | 12      | 7       |
| 9       | 22      | 2217    | 2       | 99      |
| 9       | 999     | 9       | 99      | 99      |
| 9999    | 99      | 999     | 12      | 9       |
| 6       | 7904    | 5628    | 5       | 2       |
| 9       | 999     | 9       | 99      | 12796   |
| 2022    | 9       | 999     | 16      | 16      |
| 18      | 4       | 99      | 99      | 2022    |
| 12      |         |         |         |         |
| +-----+ | +-----+ | +-----+ | +-----+ | +-----+ |
| -----+  | -----+  | -----+  | -----+  | -----+  |
| -----+  | -----+  | -----+  | -----+  | -----+  |
| -----+  | -----+  | -----+  | -----+  | -----+  |
| -----+  | -----+  | -----+  | -----+  | -----+  |
| +-----+ | +-----+ | +-----+ | +-----+ | +-----+ |
| -----+  | -----+  | -----+  | -----+  | -----+  |
| -----+  | -----+  | -----+  | -----+  | -----+  |
| -----+  | -----+  | -----+  | -----+  | -----+  |
| -----+  | -----+  | -----+  | -----+  | -----+  |
| --+     |         |         |         |         |

|                                 |                    |  |                       |  |                      |                |                    |
|---------------------------------|--------------------|--|-----------------------|--|----------------------|----------------|--------------------|
| +-----+-----+-----+-----+-----+ |                    |  |                       |  |                      |                |                    |
| summary                         | anio               |  | mes                   |  | departamento         | tipo_accidente |                    |
| +-----+-----+-----+-----+-----+ |                    |  |                       |  |                      |                |                    |
| count                           | 90891              |  | 108941                |  | 108941               |                | 9060               |
| mean                            | 2018.8934217909364 |  | 6.5385667471383595    |  | 7.878760062786279    |                | 2.894812362030905  |
| stddev                          | 2.4767653426914054 |  | 3.5336006947771166    |  | 6.966458668685116    |                | 3.4561699202772527 |
| min                             | 2015               |  | 1                     |  | 1                    |                | 1                  |
| max                             | 2022               |  | 12                    |  | 9                    |                | 99                 |
| +-----+-----+-----+-----+-----+ |                    |  |                       |  |                      |                |                    |
| +-----+-----+-----+-----+-----+ |                    |  |                       |  |                      |                |                    |
| +-----+-----+-----+-----+-----+ |                    |  |                       |  |                      |                |                    |
| +-----+-----+-----+-----+-----+ |                    |  |                       |  |                      |                |                    |
| +-----+-----+-----+-----+-----+ |                    |  |                       |  |                      |                |                    |
| +-----+-----+-----+-----+-----+ |                    |  |                       |  |                      |                |                    |
| +-----+-----+-----+-----+-----+ |                    |  |                       |  |                      |                |                    |
| +-----+-----+-----+-----+-----+ |                    |  |                       |  |                      |                |                    |
| +-----+-----+-----+-----+-----+ |                    |  |                       |  |                      |                |                    |
| +-----+-----+-----+-----+-----+ |                    |  |                       |  |                      |                |                    |
| +-----+-----+-----+-----+-----+ |                    |  |                       |  |                      |                |                    |
| +-----+-----+-----+-----+-----+ |                    |  |                       |  |                      |                |                    |
| summary                         | num_hecho          |  | dia_ocu               |  | mes_ocu              |                | dia_sem_ocu        |
| hora_ocu                        | g_hora             |  | depto_ocu             |  | mupio_ocu            |                | areag_ocu          |
| zona_ocu                        | sexo_pil           |  | edad_pil              |  | g_edad_2             |                | mayor_m            |
| enor                            | tipo_veh           |  | color_veh             |  | modelo_veh           |                | causa_acc          |
| marca_veh                       | g_edad             |  | fallecidos_lesionados |  | otro_g_edad_fall_les |                | num                |
| _correlativo                    | num_corre          |  | corre_base            |  | area_geo_ocu         |                | sexo_vic           |
| edad_vic                        | fall_les           |  | tipo_eve              |  | edad_quinquenales    |                | ano_ocu            |
| g_hora_5                        | sexo_per           |  | edad_per              |  | g_edad_80ymas        |                | g_edad_6           |
| 0ymas                           | int_o_noint        |  | g_modelo_veh          |  | filter               |                | zona_ciudad        |
| anio                            | mes                |  |                       |  |                      |                |                    |
| +-----+-----+-----+-----+-----+ |                    |  |                       |  |                      |                |                    |
| +-----+-----+-----+-----+-----+ |                    |  |                       |  |                      |                |                    |
| +-----+-----+-----+-----+-----+ |                    |  |                       |  |                      |                |                    |
| +-----+-----+-----+-----+-----+ |                    |  |                       |  |                      |                |                    |
| +-----+-----+-----+-----+-----+ |                    |  |                       |  |                      |                |                    |
| +-----+-----+-----+-----+-----+ |                    |  |                       |  |                      |                |                    |
| +-----+-----+-----+-----+-----+ |                    |  |                       |  |                      |                |                    |
| +-----+-----+-----+-----+-----+ |                    |  |                       |  |                      |                |                    |
| +-----+-----+-----+-----+-----+ |                    |  |                       |  |                      |                |                    |
| +-----+-----+-----+-----+-----+ |                    |  |                       |  |                      |                |                    |
| count                           | 9060               |  | 108941                |  | 108941               |                | 108941             |
| 108941                          | 108941             |  | 108941                |  | 108941               |                | 9060               |
| 108941                          | 9060               |  | 9060                  |  | 9060                 |                | 108941             |
| 108941                          | 108941             |  | 108941                |  | 9060                 |                | 108941             |
| 18050                           | 9060               |  | 9060                  |  | 8990                 |                | 99                 |
| 781                             | 8990               |  | 31055                 |  | 8990                 |                | 8990               |
| 99881                           | 99881              |  | 99881                 |  | 90891                |                | 90891              |
| 90891                           | 90891              |  | 90891                 |  | 90891                |                | 90891              |
| 90891                           | 11668              |  | 10544                 |  | 90891                |                | 108941             |
| +-----+-----+-----+-----+-----+ |                    |  |                       |  |                      |                |                    |
| mean                            | 4530.5             |  | 15.652968120358727    |  | 6.5385667471383595   |                | 4.358102091958032  |
| 13.588364344002716              | 2.8546552721197713 |  | 7.878760062786279     |  | 793.5396866193628    |                | 1.7214128          |
| 035320089                       | 75.11610871939857  |  | 1.3186534216335541    |  | 76.16048565121413    |                | 5.126379690949228  |
| 1.4107269072250117              | 10.146694082117843 |  | 30.420502840987325    |  | 7037.028244646184    |                | 2.89481            |
| 2362030905                      | 259.9227563543569  |  | 5.038614958448753     |  | 1.8328918322295806   |                | 3.78708609271      |

```

5232      |4495.5                |5047.8826229442475|2658.960622914349 |1.667976171308968
|1.292769744160178|78.49788654060066 |1.8212172485257456 |2.6518857440354022|7.64930
2670177511|2018.8934217909364|1.9957531548778207|1.2979282877292582|105.569506331760
02|5.759745189292669|5.349671584645344|1.2618851151379125|48.54805206236041|0.172094
61775797052|82.77551213960547|2018.8934217909364|6.5385667471383595|
|stddev |2615.5410530136974|8.830488658621803 |3.5336006947771166|2.0610779305955296
|6.6884034209718175|1.0552160543761226|6.966458668685116|696.6400467944266|0.4483286
24675187 |40.292069631232074|0.5042098488411214|205.65040599527342|1.571432490493401
8|1.447290107599545 |22.757175705645274|42.75048977963311 |4069.5912307623216|3.4561
699202772527|397.6989545665498|3.1591932933283675|0.37309300729948003 |1.5131717037
457628 |2595.3337935610516|2976.999977574603 |1833.6546635897134|0.4709470753221143
3|0.455058958155703|209.98581855390364|0.38753707700361445|2.532689893160055 |4.3033
33609598085|2.4767653426914054|0.8433749681940823|0.6122974192086612|258.59118309423
627|4.216420227466242|3.343289608693851|0.6253646629367897|46.83253287830303|0.37747
883696037926|34.42467296047124|2.4767653426914054|3.5336006947771166|
|min      |1              |1              |1              |1              |
|0          |          |1          |1          |101          |1
|0          |          |1          |1          |2            |1
|1          |          |1          |1          |1            |1
|1          |          |1          |1          |1            |1
|1          |          |1          |1          |1            |1
|1          |          |1          |1          |1            |1
|0          |          |1          |1          |1            |1
|0          |          |1          |1          |1            |
|25%       |2264          |8              |3              |3              |
|9          |          |2          |1          |110          |1
|18         |          |1          |20         |4            |1
|3          |          |2          |2014        |1            |27
|3          |          |2          |3           |2247         |24
96          |1124          |1             |1            |20           |
|2          |          |1          |5           |2017         |1
|1          |          |21         |3           |3            |1
|5          |          |0           |99          |2017         |3
|
|50%       |4529          |16            |7             |5             |
|15         |          |3          |5           |509          |2
|99         |          |1          |28          |5            |1
|4          |          |5          |9999        |3            |64
|4          |          |2          |3           |4494         |49
89          |2248          |2            |1            |28           |
|2          |          |2          |6           |2019         |2
|1          |          |29         |4           |4            |1
|6          |          |0           |99          |2019         |7
|
|75%       |6794          |23            |10            |6             |
|19         |          |4          |14          |1415         |2
|99         |          |2          |42          |6            |1
|4          |          |99         |9999        |5            |99
|7          |          |2          |4           |6742         |74
84          |4169          |2            |2            |43           |
|2          |          |5          |9           |2021         |3
|2          |          |44         |7           |7            |1
|99         |          |0           |99          |2021         |10
|
|max        |9060          |31            |12            |7             |
|99         |          |5          |22          |2217         |2

```

|         |         |         |         |         |
|---------|---------|---------|---------|---------|
| 99      | 9       | 999     | 9       | 9       |
| 99      | 99      | 9999    | 99      | 999     |
| 12      | 2       | 8       | 8990    | 11      |
| 668     | 6417    | 2       | 2       | 999     |
| 9       | 99      | 18      | 2022    | 4       |
| 9       | 999     | 16      | 12      | 9       |
| 99      | 1       | 99      | 2022    | 12      |
|         |         |         |         |         |
| +-----+ | +-----+ | +-----+ | +-----+ | +-----+ |
| +-----+ | +-----+ | +-----+ | +-----+ | +-----+ |
| -----+  | -----+  | -----+  | -----+  | -----+  |
| +-----+ | +-----+ | +-----+ | +-----+ | +-----+ |
| -----+  | -----+  | -----+  | -----+  | -----+  |
| +-----+ | +-----+ | +-----+ | +-----+ | +-----+ |
| -----+  | -----+  | -----+  | -----+  | -----+  |
| +-----+ | +-----+ | +-----+ | +-----+ | +-----+ |
| -----+  | -----+  | -----+  | -----+  | -----+  |
| +-----+ | +-----+ | +-----+ | +-----+ | +-----+ |
| -----+  | -----+  | -----+  | -----+  | -----+  |

## Años y verificación

```
In [ ]: from pyspark.sql import functions as F

years_by_ds = {}
nulls_by_ds = {}

for k, df in sdf.items():
    if "anio" in df.columns:
        ys = [r["anio"] for r in df.select("anio")
               .where(F.col("anio").isNotNull())
               .distinct().orderBy("anio").collect()]

        years_by_ds[k] = ys

        nulls_by_ds[k] = df.filter(F.col("anio").isNull()).count()
        print(f"{k}: {ys} (filas con anio nulo = {nulls_by_ds[k]})")
    else:
        print(f"{k}: (sin columna 'anio')")

if years_by_ds:
    sets = list(map(set, years_by_ds.values()))
    inter = set.intersection(*sets) if len(sets) > 1 else sets[0]
    union = set.union(*sets)
    print("\nIntersección de años:", sorted(inter))
    print("Unión de años:", sorted(union))
    if all(set(ys) == inter for ys in years_by_ds.values()):
        print("✓ Todos los datasets tienen exactamente los mismos años (ignorando nulos)")
    else:
        print("⚠ Hay diferencias de años entre datasets (ignorando nulos).")
```

hechos: [2015, 2016, 2017, 2018, 2019, 2020, 2021, 2022] (filas con anio nulo = 11975)

vehiculos: [2016, 2017, 2018, 2019, 2020, 2021, 2022] (filas con anio nulo = 24050)

fallecidos: [2015, 2016, 2017, 2018, 2019, 2020, 2021, 2022] (filas con anio nulo = 18050)

Intersección de años: [2016, 2017, 2018, 2019, 2020, 2021, 2022]

Unión de años: [2015, 2016, 2017, 2018, 2019, 2020, 2021, 2022]

⚠ Hay diferencias de años entre datasets (ignorando nulos).

## Valores distintos por accidente

```
In [14]: for k, df in sdf.items():
         if "tipo_accidente" in df.columns:
             print(f"\n{k}: tipos de accidente (distinct)")
             df.select("tipo_accidente").distinct().orderBy("tipo_accidente").show(200,
```

hechos: tipos de accidente (distinct)

```
+-----+
|tipo_accidente|
+-----+
|NULL          |
|1             |
|2             |
|3             |
|4             |
|5             |
|99            |
+-----+
```

vehiculos: tipos de accidente (distinct)

```
+-----+
|tipo_accidente|
+-----+
|NULL          |
|1             |
|2             |
|3             |
|4             |
|99            |
+-----+
```

fallecidos: tipos de accidente (distinct)

```
+-----+
|tipo_accidente|
+-----+
|NULL          |
|1             |
|2             |
|3             |
|4             |
|5             |
|99            |
+-----+
```

## Departamentos unicos por base de datos

```
In [ ]: for k, df in sdf.items():
        if "departamento" in df.columns:
            cnt = df.select("departamento").distinct().count()
            print(f"{k}: departamentos únicos = {cnt}")
            df.select("departamento").distinct().orderBy("departamento").show(50, trunc
```

hechos: departamentos únicos = 22

| +-----+      |  |
|--------------|--|
| departamento |  |
| +-----+      |  |
| 1            |  |
| 10           |  |
| 11           |  |
| 12           |  |
| 13           |  |
| 14           |  |
| 15           |  |
| 16           |  |
| 17           |  |
| 18           |  |
| 19           |  |
| 2            |  |
| 20           |  |
| 21           |  |
| 22           |  |
| 3            |  |
| 4            |  |
| 5            |  |
| 6            |  |
| 7            |  |
| 8            |  |
| 9            |  |
| +-----+      |  |

vehiculos: departamentos únicos = 22

| +-----+      |  |
|--------------|--|
| departamento |  |
| +-----+      |  |
| 1            |  |
| 10           |  |
| 11           |  |
| 12           |  |
| 13           |  |
| 14           |  |
| 15           |  |
| 16           |  |
| 17           |  |
| 18           |  |
| 19           |  |
| 2            |  |
| 20           |  |
| 21           |  |
| 22           |  |
| 3            |  |
| 4            |  |
| 5            |  |
| 6            |  |
| 7            |  |
| 8            |  |
| 9            |  |
| +-----+      |  |

```
fallecidos: departamentos únicos = 22
```

```
+-----+  
|departamento|  
+-----+  
|1             |  
|10            |  
|11            |  
|12            |  
|13            |  
|14            |  
|15            |  
|16            |  
|17            |  
|18            |  
|19            |  
|2             |  
|20            |  
|21            |  
|22            |  
|3             |  
|4             |  
|5             |  
|6             |  
|7             |  
|8             |  
|9             |  
+-----+
```